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Graph Theory

End-Sem Assignment

Answer-1

Note:- Symbols used have usual meaning.

Claim:- In a loopless graph G, $\alpha_0(G^c) = \omega(G)$

Proof:- I). $\alpha_0(G^c) \geq \omega(G)$

→ Consider $V(H)$, where $H \subseteq G$ is a maximum clique in G. In G^c, for any 2 vertices u & v in H, there'd be no edge & so $V(H)$ is an independent set in G^c.

∴ The claim holds.

(I).

II). $\omega(G) \geq \alpha_0(G^c)$

→ Assume to contrary that $\alpha_0(G^c) > \omega(G)$ & let $S = \{u \in V(G) \mid \forall v \in V(G),\ (u,v) \notin G^c\}$ be a maximum independent set in G^c.

Again, there'd be an edge b/w any two vertices of S in G ∴ $[S]$ is a complete subgraph of G with $|[S]| = \alpha_0(G^c) > \omega(G)$; which is a contradiction to the maximality of $\omega(G)$.

∴ The claim holds.

using (I). & (II). , we've :- $\alpha_0(G^c) = \omega(G)$

1(a):

Now, the given claim is wrong. Counter example :- $G = K_n$ with $\beta_0(G) + \beta_0(G^c) = n-1 < \alpha_0(G) + \alpha_0(G^c) = n+1$

Answer-1(b)

Counter example to given claim :- $G = K_n$

for it, $\omega(G) = n$

& $\beta_0(G) = n-1$

Correct claim :- $\beta_0(G) \geq \omega(G) - 1$

Proof :- Assume to contrary that S' is a minimum vertex cover in G with $|S| < \omega(G) - 1$.

Then, consider the maximum clique of $G \to H$.

let u, v be any 2 of its vertices. Then $\because (u,v) \in G$ (of H)

$\therefore$ either $u \in S$ or $v \in S$.

i.e. S is also a minimum vertex cover of G with $|S| < \omega(G) - 1$; a contradiction since $\beta_0(H) = |V(H)| - 1$, where H is a complete graph.

$\therefore$ Proof is complete.

Answer-1(c).

Let 'S' be a minimum vertex cover of G.

If $S = V(G)$, then $\beta_0(G) = |V(G)| \geq \delta(G)$ & the inference is trivial.

Else, let $v \in V(G) - S$ be any arbitrary vertex.

Then, we've:- $\forall u \mid u \in N_G(v)$, $u \in S$, where

$$N_G(v) = \{u \in V(G) \mid (u,v) \text{ is an edge in } G\}.$$

$\therefore \beta_0(G) \geq \deg(v) \geq \delta(G)$.

Hence, proved.

Answer-2

Given, M & N are two disjoint matchings of a graph G with $|M| > |N|$.

Let H be the induced subgraph of G due to edges in $M \cup N$.

In H, $\because$ any $v \in V(H)$ $(= V(G))$ can be incident on atmost two edges; $\Delta(H) \leq 2$.

Therefore, the only possible components of H are even cycles, containing alternating edges from M & N

(odd cycles would force 2 edges to be incident on a vertex from either M or N → a contradiction $\because$ M & N are matchings.)

& paths, again with alternating edges from M & N.

Each of these would be vertex disjoint components or otherwise, $\Delta(H) > 2$ → a contradiction to established result that $\Delta(H) \leq 2$.

Now, for any even cycle or even length path, X;

$|X \cap M| = |X \cap N|$

∴ for $|M| > |N|$, as given there must be an odd length path with first & last edges from M & with remaining edges alternating b/w M & N.

This would be an N-alternating path ∵ the end vertices are N-unsaturated & all edges of the form (v_{2i}, v_{2i+1}) on path belong to N.

Define : $\mathcal{A}_1 = \{(v_1, v_2), (v_3, v_4), \ldots, (v_{2k-1}, v_{2k})\}$

$\mathcal{A}_2 = \{(v_2, v_3), (v_4, v_5), \ldots (v_{2n-2}, v_{2n-1})\}$,

where v_i's are vertices on above N-alternating path.

Then, $M' = M - \mathcal{A}_1 \cup \mathcal{A}_2$ & $N' = N - \mathcal{A}_2 \cup \mathcal{A}_1$ are matchings of graph G_2,

with $|M'| = |M| - 1$ $\{\because |S_2| = |S_1| - 1\}$

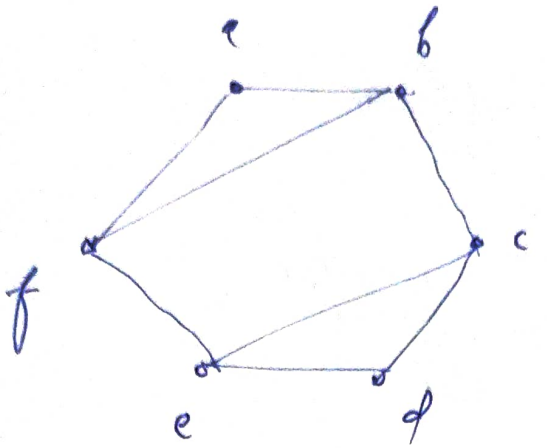
& $|N'| = |N| + 1$

& $M' \cap N' = \emptyset$

& $M' \cup N' = M \cup N$. $\{\because$ All the edges of $M \cup N$ have been assigned exactly 1 : either M' or

Answer-3

Given graph:-



(G)

Iteration	Chosen Vertex	Independent Set
1.	a	{a}
	$G_1 \leftarrow G - \{a, b, f\}$ (Neighbours of chosen vertex)	
2.	d	{a, d}

$G_2 \leftarrow G - \{d, e, c\}$

i.e. $V(G) = \phi$

∴ Stop.

Therefore, the independent set obtained at the termination of the greedy algorithm, i.e. {a, d} is a maximal independent set in G.

Answer-4

Answer-4(a).

Iteration	Chosen edge	Vertex set
1.	$(1,2)$	$\{1,2\}$
	$G_1 \leftarrow G_2 - \{1,2\}$	
2.	$(3,4)$	$\{1,2,3,4\}$
	$G_2 \leftarrow G_1 - \{3,4\}$	
3.	$(5,6)$	$\{1,2,3,4,5,6\}$
	$G_2 \leftarrow G_1 - \{5,6\}$	
	$\because E(G_1) = \phi$	
	stop.	

$\therefore$ The algorithm on termination yields :-

$\{(1,2),(3,4),(5,6)\}$ as a maximal matching

& $\{1,2,3,4,5,6\}$ as a vertex cover of graph G.

Answer 4 (b).

Iteration	Chosen edge	Vertex set
1.	$(4,5)$	$\{4,5\}$
	$G \leftarrow G - \{4,5\}$	
2.	$(2,6)$	$\{4,5,2,6\}$
	$G \leftarrow G - \{2,6\}$	
3.	$(8,10)$	$\{4,5,2,6,8,10\}$
	$G \leftarrow G - \{8,10\}$	
4.	$(3,7)$	$\{4,5,2,8,10,3,7\}$
	$G \leftarrow G - \{3,7\}$	

$\because E(G) = \phi$

$\therefore$ Stop.

$\therefore$ The algorithm on termination yields :-

$\{(2,6), (8,10), (3,7), (4,5)\}$ as maximal matching

& $\{0,2,3,4,5,6,7,8,10\}$ as a vertex cover.

Answer-5

∵ The given graph is a bipartite graph & ordering as per Brooks' theorem would allow us to achieve $\chi_0(G) = \Delta(G) = 3$.

But, to colour using 4 colours, we fix a vertex & have its neighbours coloured 1, 2 & 3, each with one.

∴ Consider the ordering of vertices as :-

$$O = \{u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4\}$$

Now, following is the dry run of the greedy algorithm :-

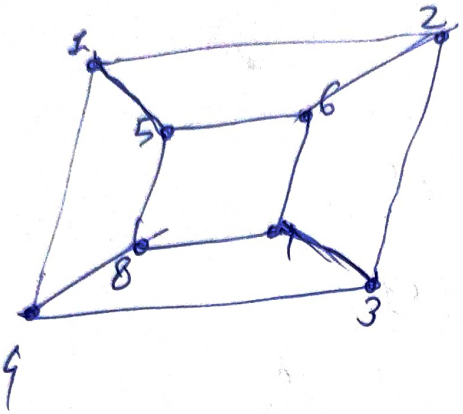
> T_i is the set of colours assigned to chosen vertex' neighbour, S_i used there & afterwards is the set of all colours assigned to vertices before i^{th} step.

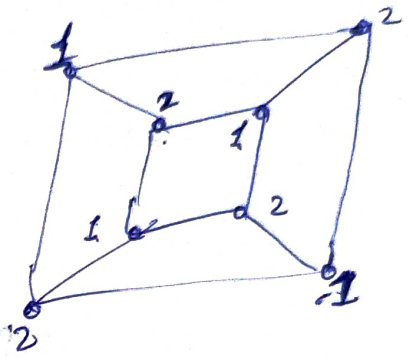
Iteration	Vertex & assigned colour	
1.	$u_1 \leftarrow 1$	$\because T_1 = \phi$
2.	$v_1 \leftarrow 1$	$\because T_2 = \phi$
3.	$u_2 \leftarrow 2$	$\because T_3 = \{1\} \Rightarrow S_i = \{1\}$
4.	$v_2 \leftarrow 2$	$\because \min\{S_i - T_4\} = 2$
5.	$u_3 \leftarrow 3$	$\because T_5 = \{1,2\} \Rightarrow S_i = \{1,2\}$
6.	$v_3 \leftarrow 3$	$\because T_6 = S_i = \{1,2,3\} - \{1,2\} = \{3\}$ & $\min\{S_i - T_6\} = 3$
7.	$u_4 \leftarrow 4$	$\because T_6 = \{1,2,3\} \Rightarrow S_i = \{1,2,3\}$
8.	$v_4 \leftarrow 4$	$\because \min\{S_i - T_8\} = 4$.

Therefore, the greedy algorithm does achieve a 4-vertex coloring of given graph with the claimed ordering of vertices.

Answer 6

Q_3 :



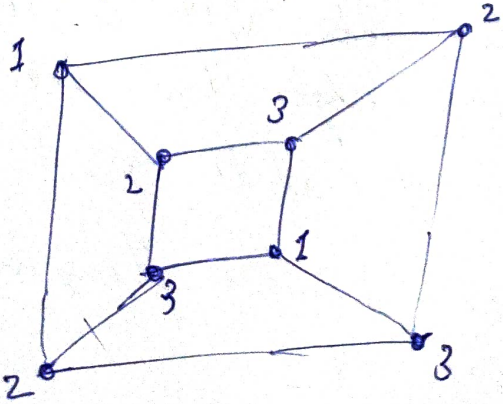


Answer - 6(a).

Considering the ordering $O_Q = \{1, 3, 6, 8, 5, 7, 2, 4\}$ inspired by the bipartite structure of Q_3 achieves a 2-coloring of Q_3 using the greedy algorithm. as shown above.

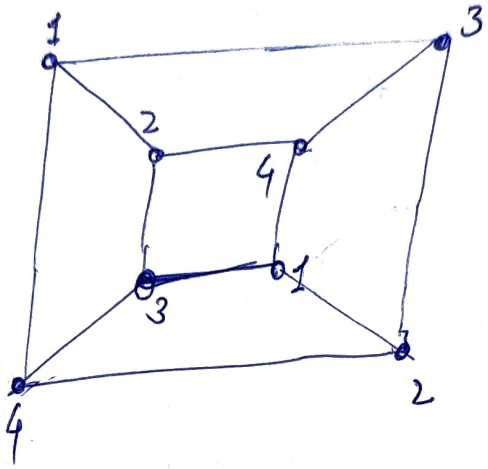
Answer - 6(b).

Consider the ordering $O_Q = \{1, 7, 2, 5, 6, 8, 4, 3\}$ & running the greedy algorithm with it as input achieves its 3-coloring as follows:-



Answer-6(a).

Consider the ordering $\sigma_2 = \{1, 7, 5, 3, 2, 6, 8, 4\}$ & running the greedy algorithm with it as input achieves a 4-colouring of G_2 as follows:-



Answer-7

Answer-7(a).

Claim:- $G_k = T_k + C_{2k+1}$, where T_k is a tree on k vertices can be coloured in $(\Delta(T) + 2k+1)$ colours. (edge colouring); where $\Delta(T)$ is the max. degree vertex of T.

Proof:- (I). $\chi_1(G) \leq \Delta(T) + 2k+1$.

Consider the vertex sets $V_1(T)$ of tree & $V_2(C_{2k+1})$; along with the cross-edges b/w these, i.e-

$[V_1, V_2]$ is a bipartite graph

& therefore using König's theorem;

$\chi_1([V_1, V_2]) = 2k+1$

, This is also readily inferred by the colour scheme :-

$c(u_{ij}) = (i+j) \% (2k+1)$, where u_{ij} is the edge joining i^{th} vertex on V_1, & j^{th} vertex in V_2.

$\Delta\ i \in \{0, 1, \ldots, k-1\}$

$\Delta\ j \in \{0, 1, \ldots, 2k\}$

Again, using König's theorem, $\because T$ is also a bipartite graph ; $\chi_1(T) = \Delta(T)$

Now, the colour sets corresponding to min. edge colouring of T & of $[V_1, V_2]$ can't have anything in common.

→ Again, consider an edge in G_{2k+1}, the size of union of colours present at its ends is atmost $2k$ & so that edge can be given the rem. colour from the colours in $[V_1, V_2]$.

$$\therefore \chi_1(G) \leq \Delta(T) + \Delta([V_1, V_2])$$
$$= \Delta(T) + 2k+1$$

(II). $\chi_1(G) \geq \Delta(T) + 2k+1$

$\Rightarrow \because \Delta(G) = \Delta(T) + 2k+1$ (for any vertex in T')

$\therefore$ Inference is trivial using Gupta-Vizing theorem

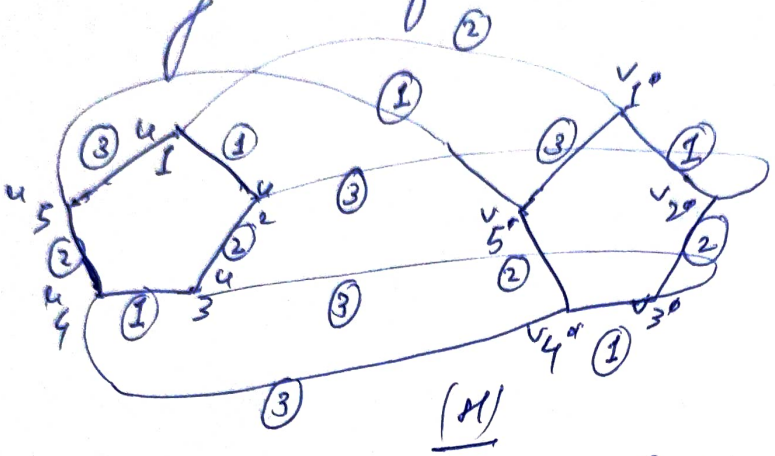
→ Now, using (I). & (II). together establishes the initial claim.

Answer-7(b).

(Claim:- $G = C_5 \square C_5$ can be edge-coloured using a min. of 7 colours.

Proof:- (I). $\chi_1(G) \leq 7$

We outline a partitioning of $E(G)$ into matchings as follows:-



(H)

→ In the above spanning subgraph H of G, the edge sets have been partitioned into 3 matchings, with membership clear from labels.

→ The only other edges are the remaining cross edges b/w the 2 cycles: Consider the spanning subgraph with now only those edges: $G-H$

Clearly, $G-H$ is a 4-regular bipartite graph $E(G)$ of which can be partitioned into 4 matchings;

with j^{th} matching having edges of the from $u_i v_{(i+j) \bmod 5}$

where, $i \in \{0,1,2,3,4\}$ & $j \in \{1,2,3,4\}$ { after assigning $u_i u_{i+1}$ the number $(i-1)$ }

Moreover, each of the now available 7 matchings (3 from H & 4 from $G-H$) can be given a distinct colour;

$\therefore \chi_1(G) \leq 7$

(II) $\chi_1(G) \geq 7$

$\because \Delta(G) = 7$

$\therefore$ The inference is trivial using Vizing theorem

→ Therefore, using (I). & (II). ; the original claim is established.

Answer-7(c).

Note: $G_7 = C_{2s} \cup [V_1; V_2] \cup C_{2t}$, where

$V_1 : V(C_{2s})$ & $V_2 : V(C_{2t})$

Claim:- $\chi_1(G) = \max(2s, 2t) + 2$

Proof:- Now; (I): $[V_1, V_2]$ is a bipartite graph, it'll require $\Delta([V_1, V_2]) = \max(2s, 2t)$ min. colours for colouring edges of

& similarly C_{2s} will require 2 colours for its min. edge colouring. (Kőnig's theorem)

Moreover, the C_{2t} cycle can be coloured with colours used for min. edge colouring of C_{2s} $\because$ no two edges in these are adjacent, $\therefore \chi_1(G) \leq \max(2s, 2t) + 2$

(II). $\chi_1(G) \geq \max(2s, 2t) + 2$

$\because \Delta(G) = \max(2s, 2t) + 2$

$\therefore$ This observed immediately using the Vizing theorem.

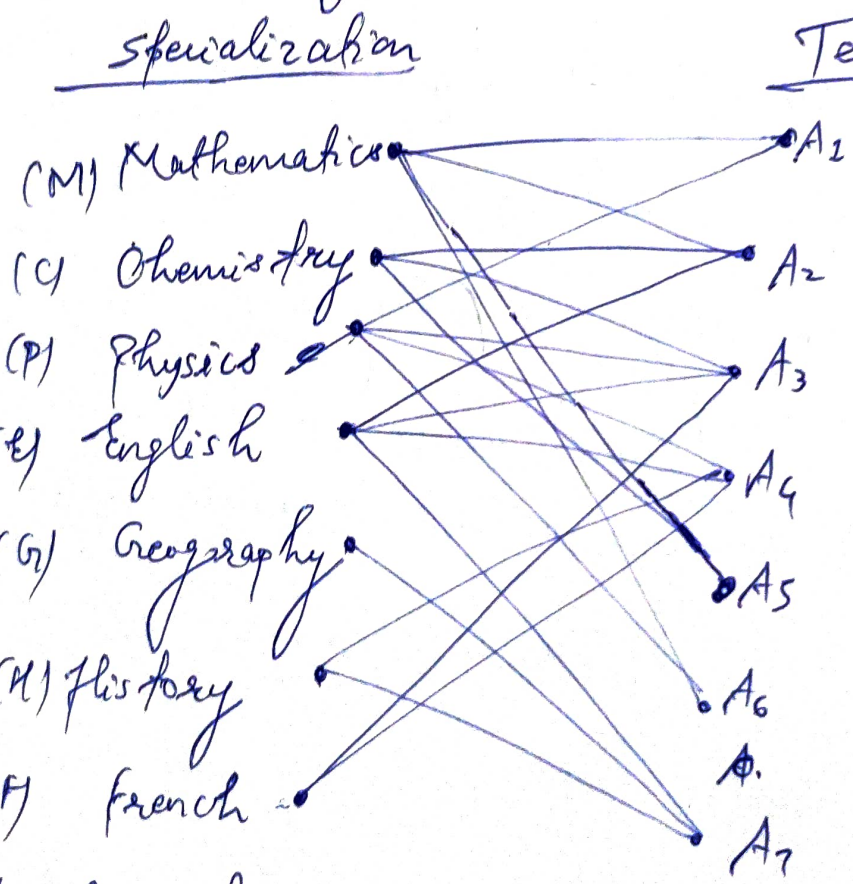
→ Therefore, using (I). & (II). completes the proof of the original claim.

Answer - 6

c. The natural constraints of the problem are to find an assignment maximising teacher-course pairings such that no two teachers are teaching the same course with each teacher allowed to teach only one course, when employed.

→ Clearly, if we model this using a bipartite graph where an edge is b/w a teacher i & a course j iff he/she can teach that course, then we're tasked to find a maximum matching of it.

→ The graph corresponding to the problem is:-



→ Upon slight observation; a perfect matching can be found as:- $M = \{ M \leftrightarrow A_6 ;\ C \leftrightarrow A_5 ;\ P \leftrightarrow A_1 ;\ E \leftrightarrow A_2 ;\ G \leftrightarrow A_7 ;\ H \leftrightarrow A_4 ;\ F \leftrightarrow A_3 \}$

→ A rigorous justification can be given by Tedious verification of Tutte's theorem or even Hall's theorem (∵ both partitions are equal in size).

Answer 9

Claim:- Any graph G satisfying $m > \alpha_1(G) \cdot \Delta(G)$ is a class-2 graph.

Proof:- We've $\Delta(G)\,\alpha_1(G) \cdot m \leq \chi_1(G) \cdot \alpha_1(G)$ { ∵ Using Pigeonhole principle: $\alpha_1(G) \geq \left\lceil \frac{m(G)}{\chi_1(G)} \right\rceil$ }

$$\Rightarrow \chi_1(G) > \Delta(G) \quad -(i).$$

Again, using Gupta-Vizing theorem :-

$$\chi_1(G) \leq \Delta(G) + 1. \quad -(ii).$$

∴ Using (i) & (ii). : $\chi_1(G) = \Delta(G) + 1$

→ Thus, the claim is proven.

Now, for n odd, $\alpha_1 \leq \lceil n/2 \rceil = \left(\frac{n-1}{2}\right)$

$$\therefore m > \left(\frac{n-1}{2}\right) \cdot \Delta(G) \geq \alpha_1(G) \cdot \Delta(G)$$

$$\Rightarrow m > \alpha_1(G) \cdot \Delta(G)$$

→ Again, using above claim ; $\chi_1(G) = \Delta(G) + 1$.

Note:- In given graph:- $\chi_1(G) = 3$

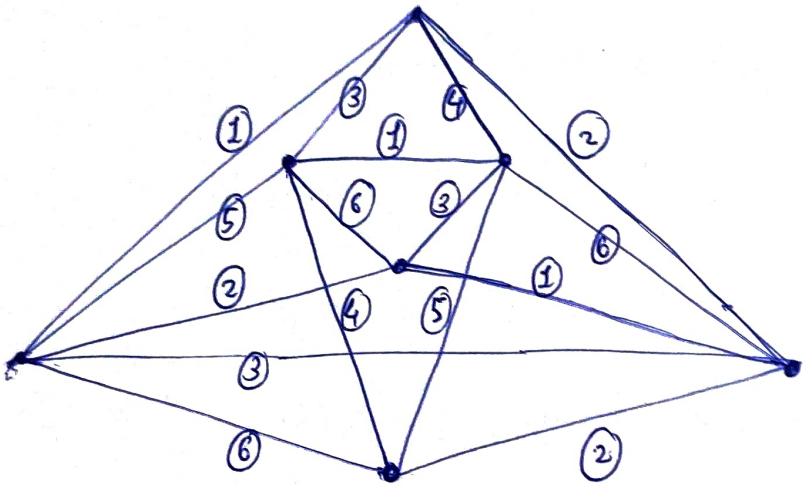
$\Delta(G) = 5$

& $m(G) = 26$.

$\therefore m(G) > \Delta(G) \cdot \chi_1(G)$

$\therefore \chi_1(G) = \Delta(G) + 1 = 6$.

A suitable min-edge colouring is as follows where colours assigned to edges are shown in circles beside them:-



Answer-10

Answer-10(a).

$\because m = \frac{\sum \deg(v)}{2} = \frac{3 \times 10}{2} = 15$ ⇒

Also; $\because 2m \geq g_0 r$, where g_0 is the girth of the plane graph if such a plane embedding of G exists.

$\Rightarrow r \leq \frac{2 \times 15}{5} = 6$

Also; for a plane graph (not necessarily connected) :-

We've:- $n - m + r = k + 1$, where k is the no. of connected components of the plane graph.

≥ 2

$\Rightarrow \quad r \geq 2 + m - n = 17 - 10 = 7$

→ A contradiction on possibilities of r.

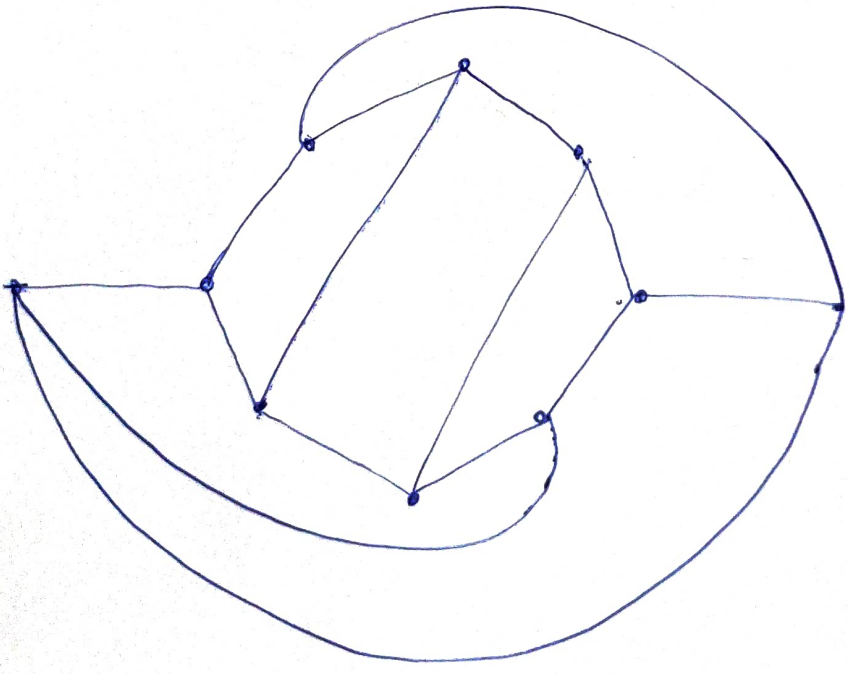
Therefore, the claim that :- "There exists no 5-regular simple planar graph on 10 vertices with girth 5" is established.

Answer - 10(6).

$\rightarrow m = \frac{\sum \deg(v_i)}{2} = \frac{3 \times 10}{2} = 15$

→ for a connected simple plane graph: $r = 2 + m - n$

$= 7$



(G)

Answer-11

Considering symbols with usual meanings:-

We've:- $2m = 3n$

$f = p + h$ ($\because 2m = \sum \deg(v_i)$)

$2m = \sum \deg(f_i) = 5p + 6h$ (f_i is face)

$n - m + f = 2$ (for a simple, plane connected 3 regular graph)

$\Rightarrow \quad \frac{2m}{3} - m + f = 2$

$\Rightarrow \quad -\frac{(5p+6h)}{6} + p + h = 2$

$\Rightarrow \quad \frac{p}{6} = 2$

$\Rightarrow \quad p = 12$

$\therefore n(G) = \frac{\chi(5p+6h)}{3\chi} = 2h + 20$

$\therefore$ Proved.

Answer-12

∵ For a simple planar graph $(n \geq 3)$; $m \leq 3n-6$.

For its complement to also be simple planar;

$$m' = \frac{n(n-1)}{2} - m \leq 3n-6$$

$$\Rightarrow \quad 2(3n-6) \geq \frac{n(n-1)}{2}$$

$$\Rightarrow \quad 4(3n-6) \geq n(n-1)$$

$$\Rightarrow \quad n^2 - 13n + 24 \leq 0$$

$$\Rightarrow \quad n = \frac{13 \pm \sqrt{169-96}}{2} = \frac{13 \pm \sqrt{73}}{2} \text{ are zeroes}$$

of polynomial in LHS.

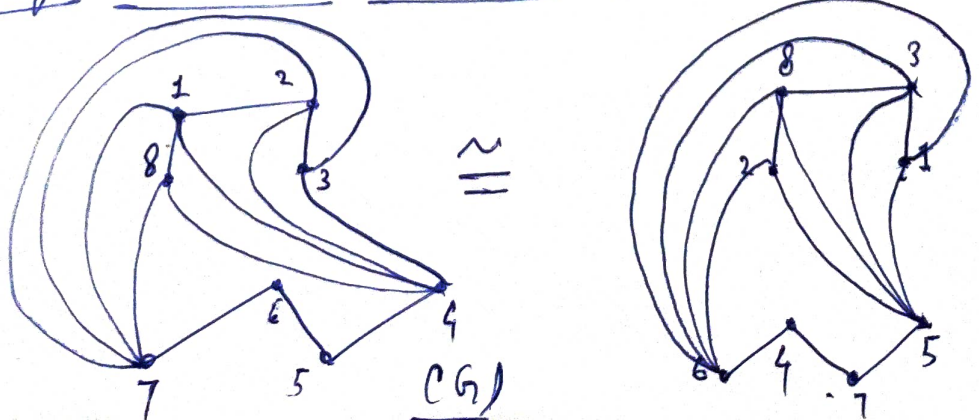
∴ Ineq. is true for $n \in \left[\frac{13-\sqrt{73}}{2}, \frac{13+\sqrt{73}}{2}\right]$

i.e. $n \leq \frac{13+8}{2} = 10$

→ Therefore; no simple planar graph with at least 11 vertices is possible such that its complement is also planar.

∴ Proved.

⇒ A self-complementary simple planar graph (n=8)



(G)